MATH 201 Your name: ____________________

Learning target DF5, version 1

Compute derivatives using the Chain Rule.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (chain, product, quotient, sum and difference, etc.) you are using.

1. $h(t) = 16\left(t + e^t + 2\right)^4$

2. $f(w) = 5\sin\left(w^{\frac{7}{3}}\right)$

3. $g(x) = 5\left(\sin(x)\right)^{\frac{7}{3}}$

4. $p(y) = 2\ln\left(\sin(y) + y^2 + 3\right)$

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Learning target DF5, version 2

Compute derivatives using the Chain Rule.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (chain, product, quotient, sum and difference, etc.) you are using.

1. $h(x) = -9\cos\left(-5x^2 + 3x + 4\right)$

2. $f(y) = -9\sin\left(y^{7/2}\right)$

3. $g(t) = -9\left(\sin(t)\right)^{7/2}$

4. $k(w) = (3w + e^w - 1)^4$

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Learning target DF5, version 3

Compute derivatives using the Chain Rule.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (chain, product, quotient, sum and difference, etc.) you are using.

1. $k(w) = -(5w + 2e^{w} + 1)^{3}$

2. $g(y) = 3\sin\left(y^{2/5}\right)$

3. $h(x) = 6\cos\left(-5x^{2} - x + 4\right)$

4. $f(t) = 3\left(\sin(t)\right)^{2/5}$

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Learning target DF5, version 4

Compute derivatives using the Chain Rule.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (chain, product, quotient, sum and difference, etc.) you are using.

1. $g(t) = 3\cos\left(-3t^2 - 3t + 4\right)$

2. $h(y) = 5\sin\left(y^{7/2}\right)$

3. $k(w) = 5\left(\sin(w)\right)^{7/2}$

4. $f(x) = -(5x + 3e^x - 4)^5$